GLAM: Efficient Continual Learning at Scale via Grouped LoRA Adapter Merging

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1. Motivation

- **Large pre-trained models** provide strong representations for adapting to new tasks.
- **LoRA** enables efficient adaptation, but existing LoRA-based approaches for continual learning still struggle to scale as tasks accumulate.
- We propose a method that **controls adapter growth, promotes knowledge reuse, and remains effective over long task sequences**.

2. Setting

- **Frozen ViT backbone** pre-trained on ImageNet.
- **Class-incremental learning** (no task IDs at test time) over long task sequences without past-task data.

3. Grouped LoRA Adapter Merging (GLAM)

- **Train a task-specific adapter.** For each task T_i, GLAM learns a new LoRA adapter ΔW_i together with the scaling factors α_i and α_{G_j} on top of the frozen backbone W_0 and the current group adapters ΔW_{G_j}, according to

$$h_x = W_0 x + \sum_{j=1}^{M} \alpha_{G_j} \Delta W_{G_j} x + \alpha_i \Delta W_i x.$$

  This makes the new adapter focus on residual task-specific information, while the scaling factors adaptively weight old and new knowledge.

- **Assign to a group.** The new adapter is randomly assigned to one of the groups.
- **Prune and merge.** The task adapter is first pruned by retaining only the top $k\%$ highest-magnitude weights to reduce redundancy and interference, and is then linearly merged into the selected group adapter, maintaining a bounded number of stored modules as tasks accumulate.
- **Align the classifier.** The classifier is updated using synthetic last-layer features sampled from stored class statistics, reducing bias toward recent classes without storing past data.
- **Build the final adapter.** All group adapters are concatenated into a single LoRA module, yielding a compact final model with efficient inference.

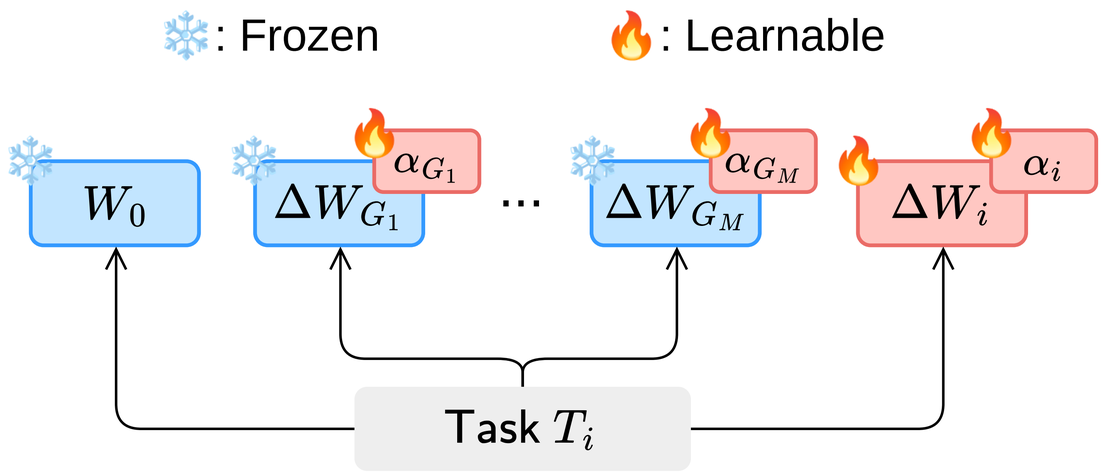


Figure 1: Task-specific LoRA training in GLAM.

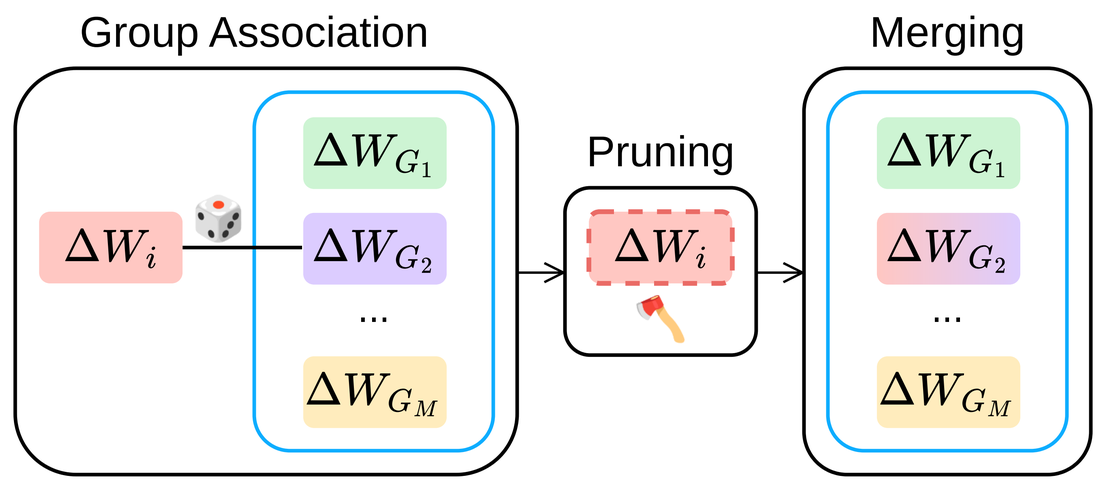


Figure 2: GLAM adapter grouping.

4. Results

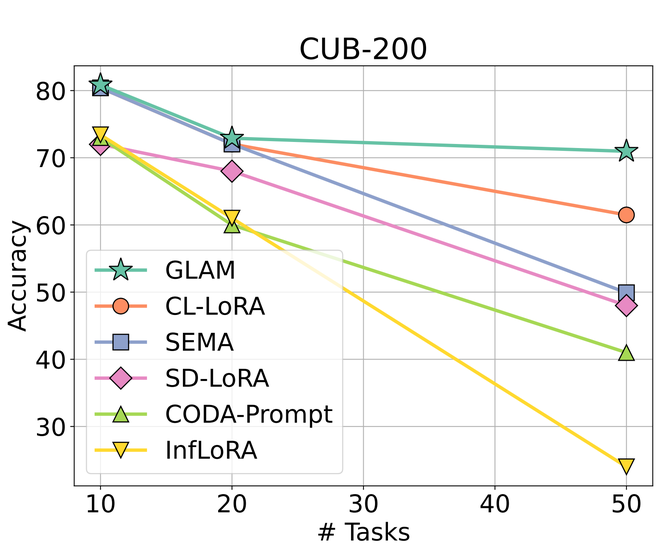


Figure 3: Final accuracy on CUB-200 with 10, 20, and 50 tasks. Performance drops more markedly for state-of-the-art methods as the number of tasks increases, while GLAM shows a more graceful degradation.

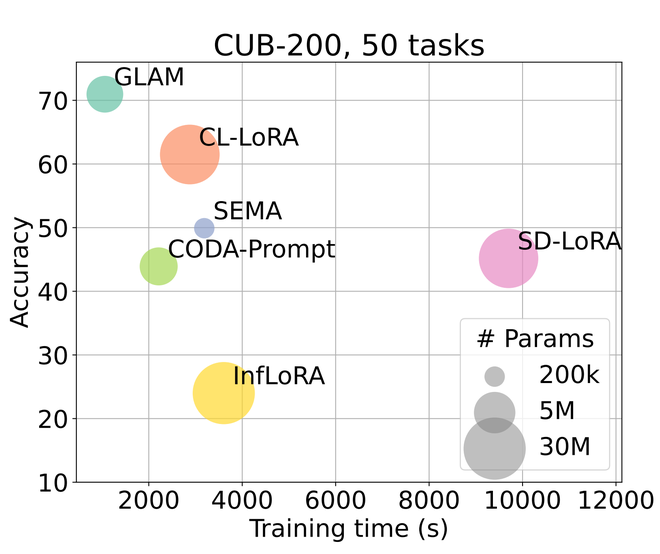


Figure 4: Efficiency–performance trade-off for 50 tasks on CUB-200. GLAM occupies the most favorable region of the plot, combining higher accuracy with lower training time and competitive parameter count relative to prior methods.

Dataset	Sim.	Orth.	Rand.
CIFAR-100	85.19 ± 0.29	84.88 ± 0.05	84.73 ± 0.74
CUB-200	69.45 ± 1.17	74.93 ± 0.95	70.96 ± 2.06
ImageNet-R	73.88 ± 0.29	71.58 ± 0.67	73.71 ± 0.68

Table 1: Final accuracy for different grouping strategies. Random grouping remains competitive with more sophisticated strategies, while being simpler and more efficient.

# Groups	ACC
1	71.71 ± 0.95
2	72.99 ± 0.87
5	73.71 ± 0.68
10	**74.23** ± 0.47
15	73.21 ± 0.84

Table 2: Final accuracy for different numbers of groups on ImageNet-R. Performance improves up to 10 groups and then saturates, while remaining strong even for small group counts.